

Задание 1

В MySQL Workbench проверьте работоспособность примеров 1-11 на созданной БД «Books».

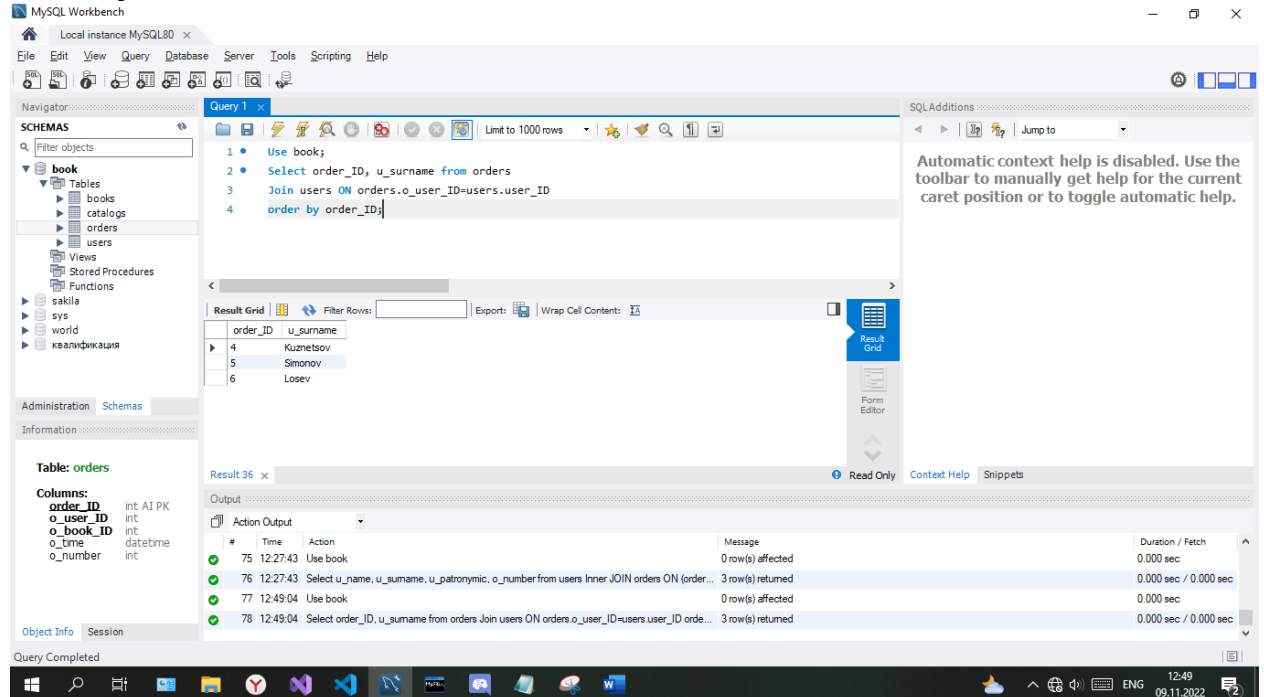
Пример 1

Use book;

Select order_ID, u_surname from orders

Join users ON orders.o_user_ID=users.user_ID

order by order_ID;



Пример 2

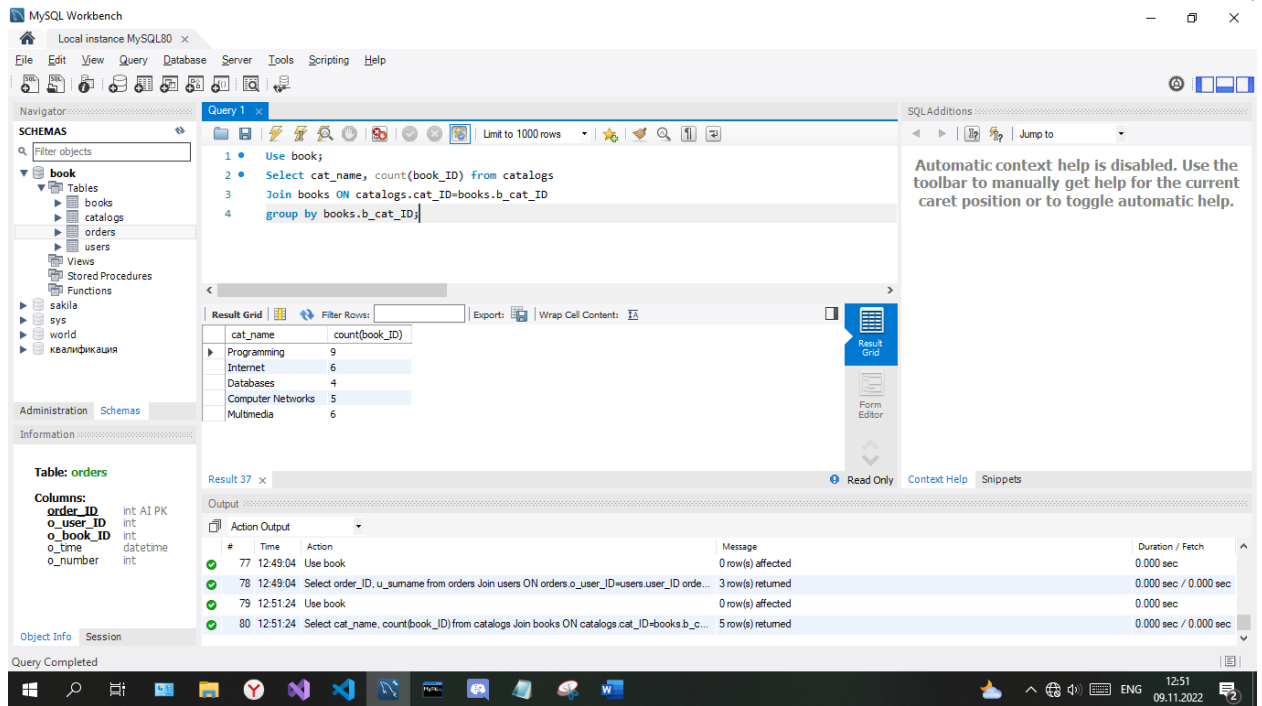
Use book;

Select cat_name, count(book_ID) from catalogs

Join books ON catalogs.cat_ID=books.b_cat_ID

group by books.b_cat_ID;

Лабораторная работа №33-34. Поддержка нескольких соединений. Создание БД с помощью средств языка PHP.

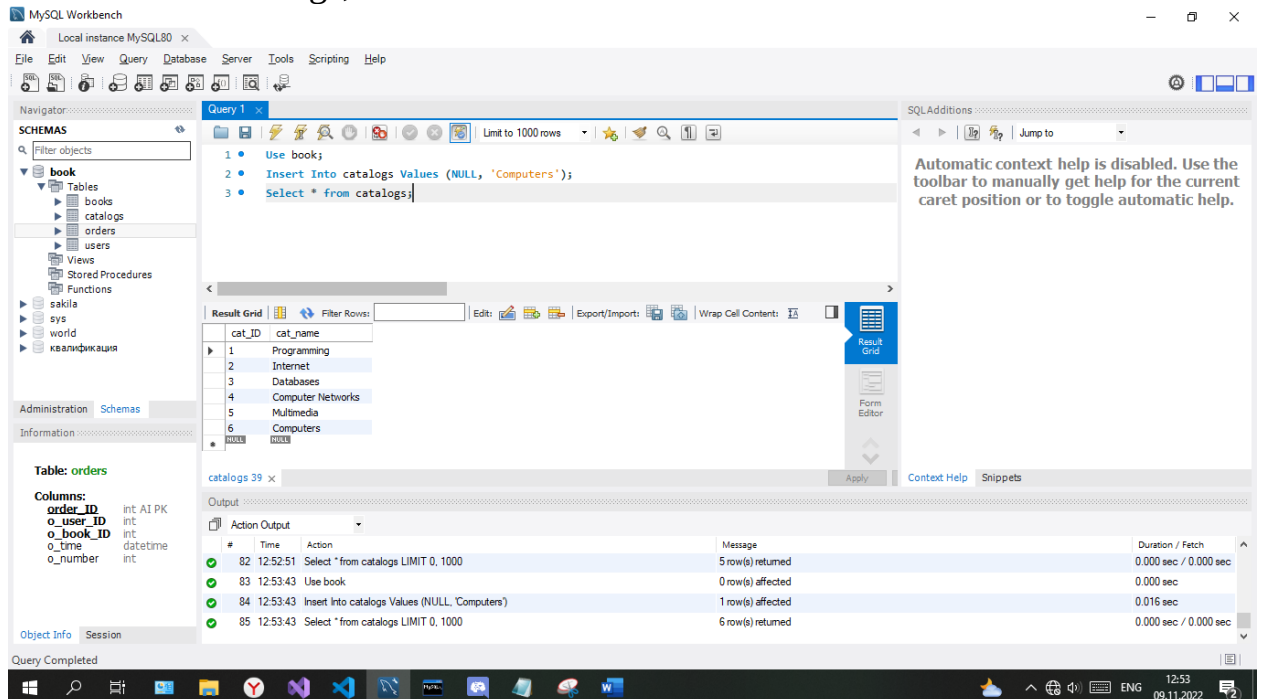


Пример 3

Use book;

Insert Into catalogs Values (NULL, 'Computers');

Select * from catalogs;



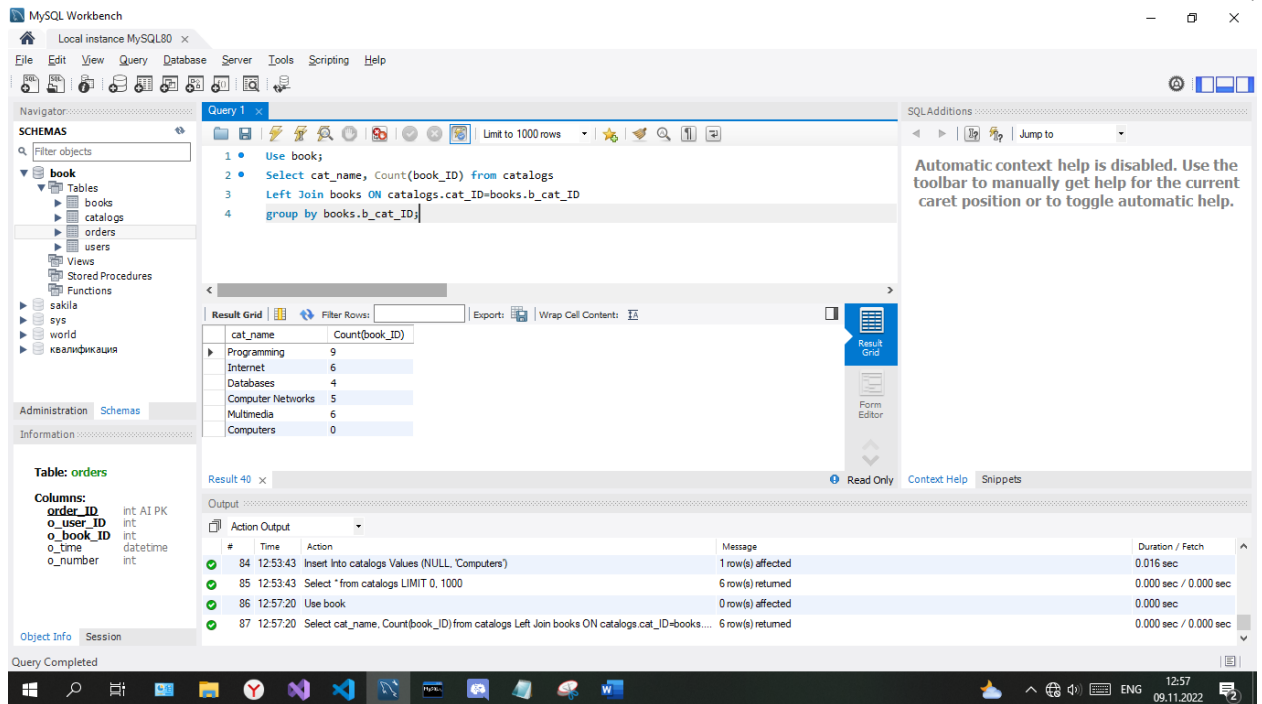
Use book;

Select cat_name, Count(book_ID) from catalogs

Left Join books ON catalogs.cat_ID=books.b_cat_ID

group by books.b_cat_ID;

Лабораторная работа №33-34. Поддержка нескольких соединений. Создание БД с помощью средств языка PHP.



Пример 4

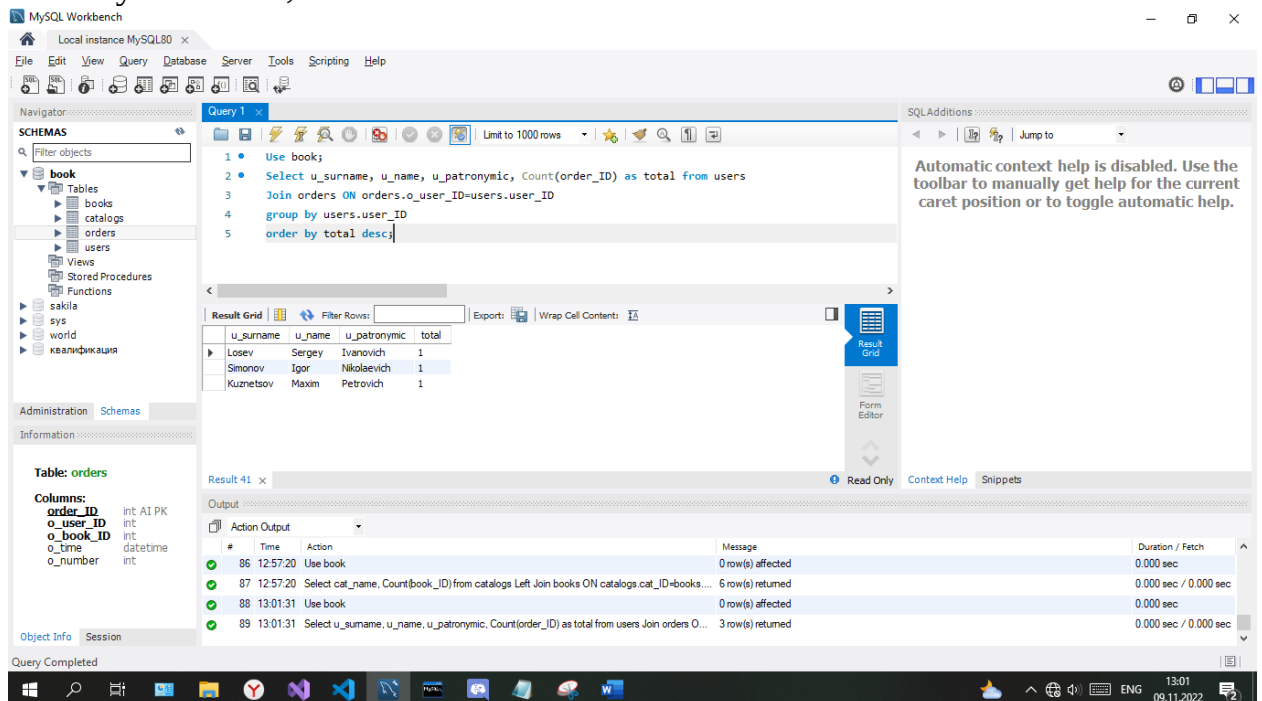
Use book;

Select u_surname, u_name, u_patronymic, Count(order_ID) as total from users

Join orders ON orders.o_user_ID=users.user_ID

group by users.user_ID

order by total desc;



Пример 5

Use book;

Select u_surname, u_name, u_patronymic, Count(order_ID) as total from users

Left Join orders ON orders.o_user_ID=users.user_ID

group by users.user_ID
order by total desc;

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query:
1 Use book;
2 Select u_surname, u_name, u_patronymic, Count(order_ID) as total from users
3 Left Join orders ON orders.o_user_ID=users.user_ID
4 group by users.user_ID
5 order by total desc;
The 'Result Grid' shows the following data:

u_surname	u_name	u_patronymic	total
Losev	Sergey	Ivanovich	1
Simonov	Igor	Nikolaevich	1
Kuznetsov	Maxim	Petrovich	1
Ivanov	Alexander	Valerevich	0
Petrov	Anatoly	Yuryevich	0
Korneev	Alexander	Alexandrovich	0

The 'Output' tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
88	13:01:31	Use book;	0 row(s) affected	0.000 sec
89	13:01:31	Select u_surname, u_name, u_patronymic, Count(order_ID) as total from users Join orders O...	3 row(s) returned	0.000 sec / 0.000 sec
90	13:03:46	Use book;	0 row(s) affected	0.000 sec
91	13:03:46	Select u_surname, u_name, u_patronymic, Count(order_ID) as total from users Left Join order...	6 row(s) returned	0.000 sec / 0.000 sec

Пример 6

Use book;

Select b_name, b_price from books

Where b_cat_ID=(Select cat_ID from catalogs where cat_name='Databases')

order by b_price;

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query:
1 Use book;
2 Select b_name, b_price from books
3 Where b_cat_ID=(Select cat_ID from catalogs where cat_name='Databases')
4 order by b_price;
The 'Result Grid' shows the following data:

b_name	b_price
Workshop on Access	74.59
Databases, Application Development	179.55
Uncovering the Secrets of SQL	190.00
Databases	309.70

The 'Output' tab shows the execution log:

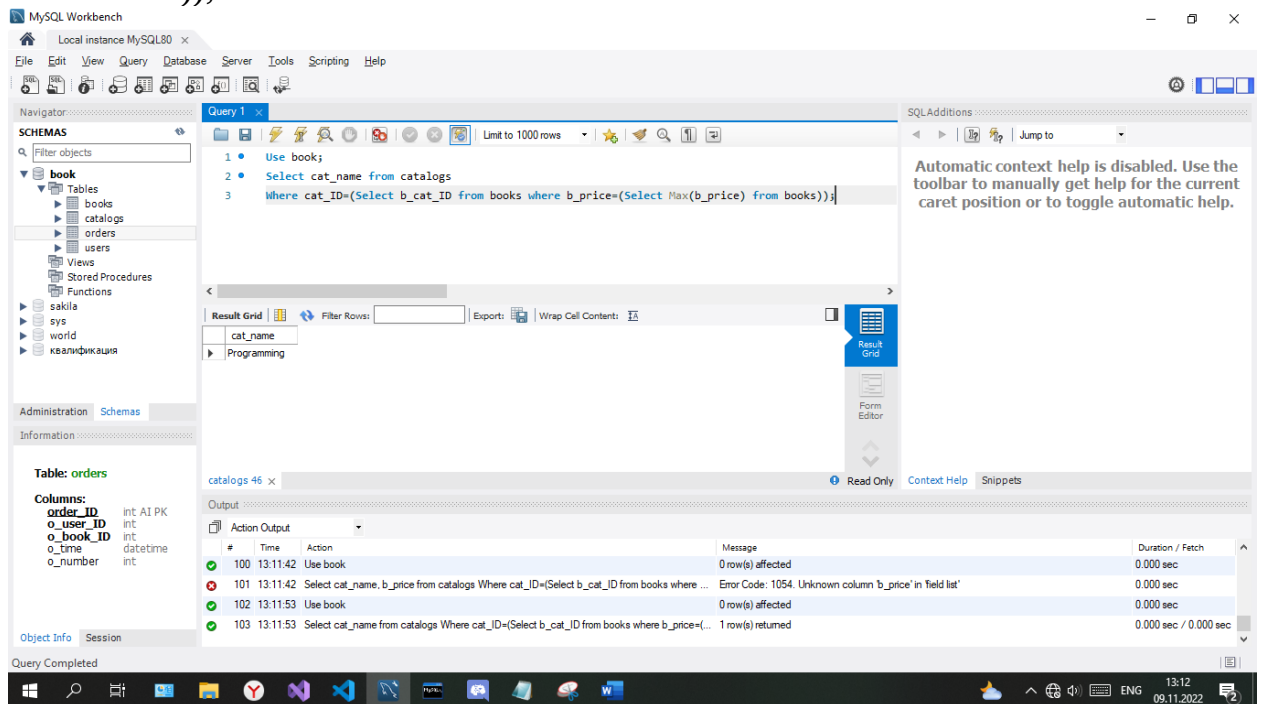
#	Time	Action	Message	Duration / Fetch
94	13:07:09	Use book;	0 row(s) affected	0.000 sec
95	13:07:09	Select * from catalogs LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
96	13:07:22	Use book;	0 row(s) affected	0.000 sec
97	13:07:22	Select b_name, b_price from books Where b_cat_ID=(Select cat_ID from catalogs where ca...	4 row(s) returned	0.000 sec / 0.000 sec

Пример 7

Use book;

Select cat_name from catalogs

Where cat_ID=(Select b_cat_ID from books where b_price=(Select Max(b_price) from books));

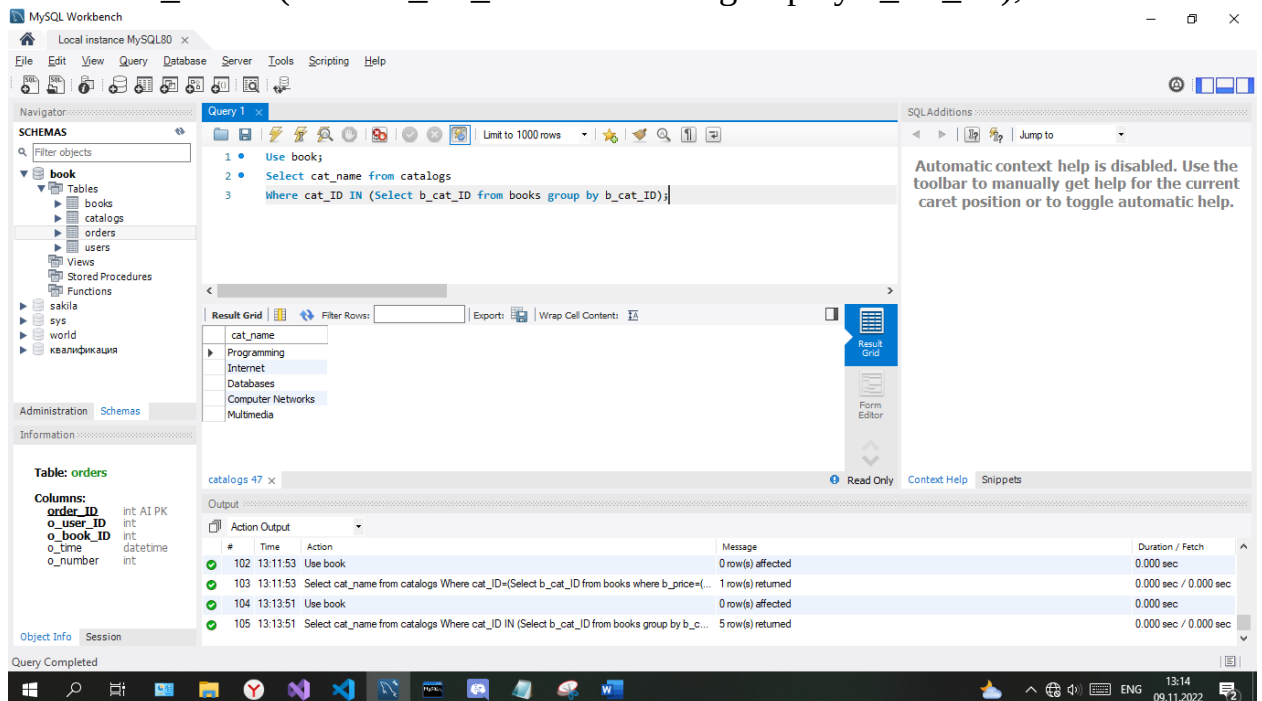


Пример 8

Use book;

Select cat_name from catalogs

Where cat_ID IN (Select b_cat_ID from books group by b_cat_ID);



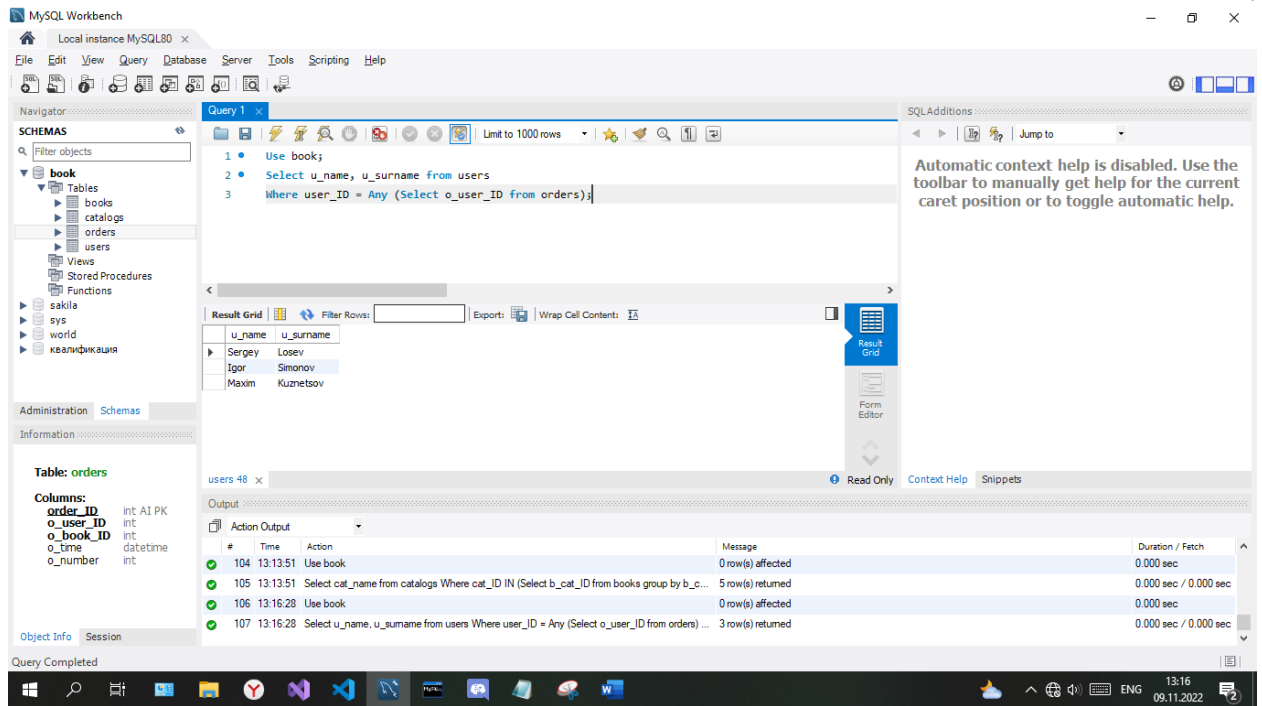
Пример 9

Use book;

Select u_name, u_surname from users

Where user_ID = Any (Select o_user_ID from orders);

Лабораторная работа №33-34. Поддержка нескольких соединений. Создание БД с помощью средств языка PHP.

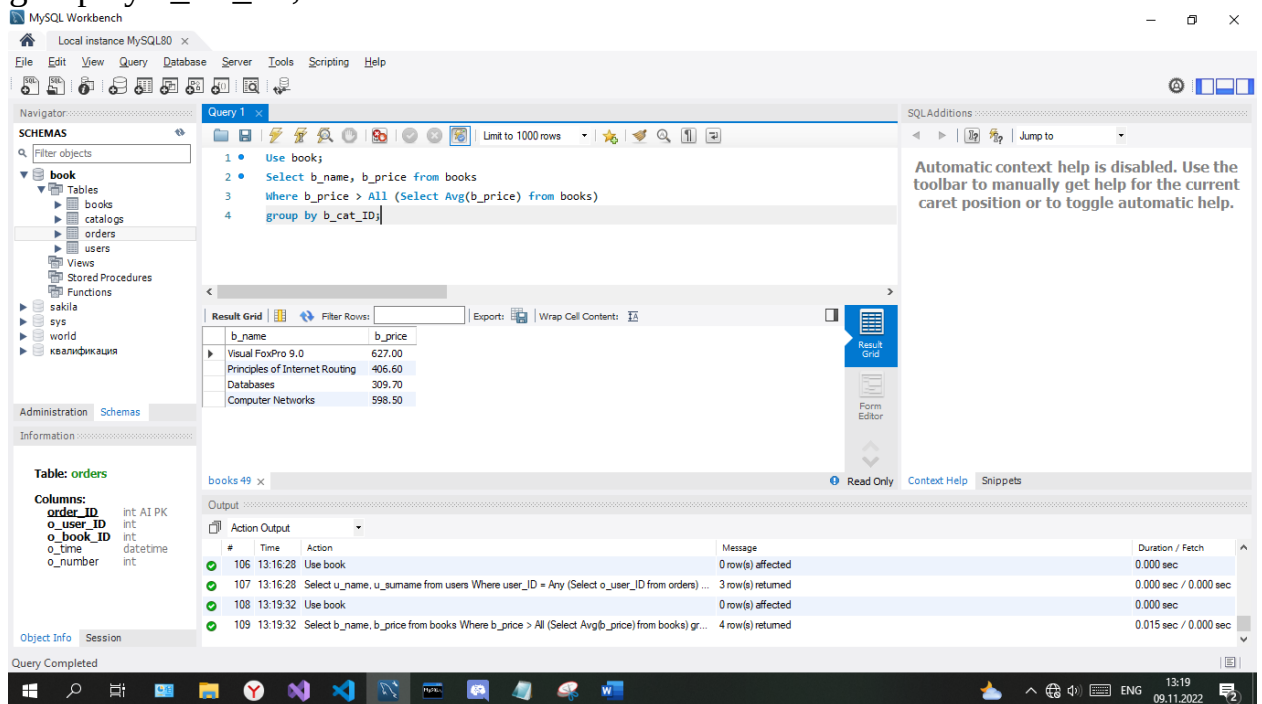


Пример 10

Use book;

Select b_name, b_price from books

Where b_price > All (Select Avg(b_price) from books)
group by b_cat_ID;



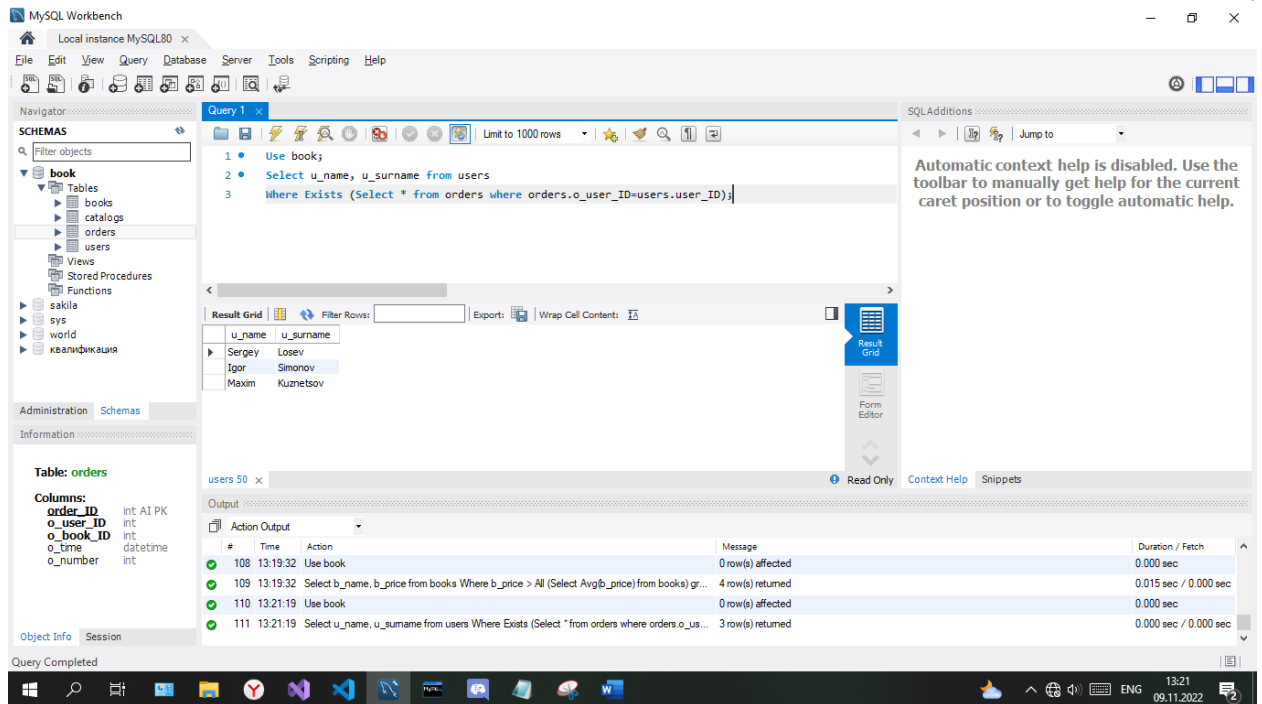
Пример 11

Use book;

Select u_name, u_surname from users

Where Exists (Select * from orders where orders.o_user_ID=users.user_ID);

Лабораторная работа №33-34. Поддержка нескольких соединений. Создание БД с помощью средств языка PHP.



Задание 2

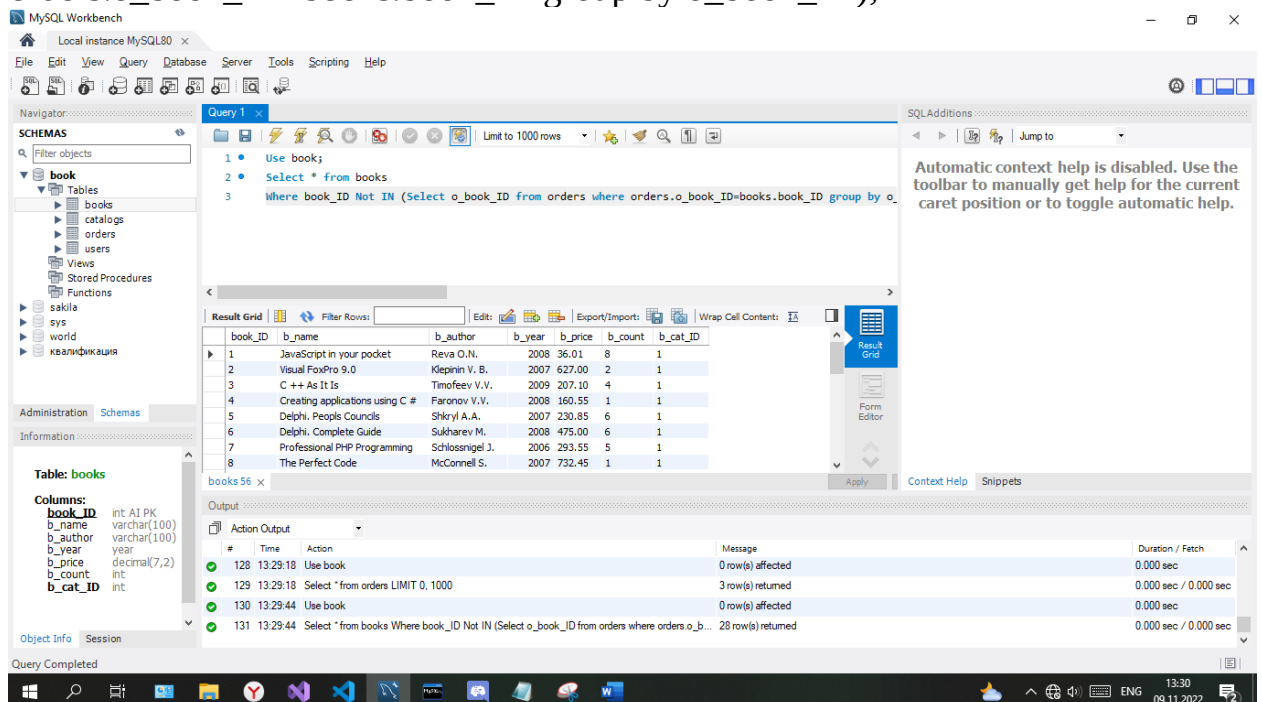
Реализуйте запросы

2.1 Создайте запрос на выборку, выводящий перечень книг, которые не заказывались пользователями

Use book;

Select * from books

Where book_ID Not IN (Select o_book_ID from orders where orders.o_book_ID=books.book_ID group by o_book_ID);



2.2 Создайте запрос на выборку, выводящий перечень каталогов, книги из которых не заказывались пользователями

Use book;

Select cat_name from catalogs

Where cat_ID Not In (Select b_cat_ID from books where books.b_cat_ID=catalogs.cat_ID);

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
1 Use book;
2 Select cat_name from catalogs
3 Where cat_ID Not In (Select b_cat_ID from books where books.b_cat_ID=catalogs.cat_ID);
```

The 'Result Grid' tab is active, showing the output of the query. The output is a table with one column, 'cat_name', and one row with the value 'Computers'.

The 'Output' tab is also visible, showing the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
144	13:39:06	Use book	0 row(s) affected	0.000 sec
145	13:39:06	Select cat_name from catalogs Where cat_ID=(Select b_cat_ID from books where books.b_cat_ID=catalogs.cat_ID)	Error Code: 1242. Subquery returns more than 1 row	0.000 sec
146	13:39:29	Use book	0 row(s) affected	0.000 sec
147	13:39:29	Select cat_name from catalogs Where cat_ID Not In (Select b_cat_ID from books where books.b_cat_ID=catalogs.cat_ID)	1 row(s) returned	0.000 sec / 0.000 sec

2.3 Вывести все данные о книгах под названием «Computer Networks», «The Perfect Code», использовать конструкцию IN

Use book;

Select * from books Where b_name In ('Computer Networks', 'The Perfect Code');

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
1 Use book;
2 Select * from books Where b_name In ('Computer Networks', 'The Perfect Code');
```

The 'Result Grid' tab is active, showing the output of the query. The output is a table with columns: book_ID, b_name, b_author, b_year, b_price, b_count, b_cat_ID. The output shows two rows:

book_ID	b_name	b_author	b_year	b_price	b_count	b_cat_ID
8	The Perfect Code	McConnell S.	2007	732.45	1	1
20	Computer Networks	Tannenbaum E.	2007	598.50	6	4

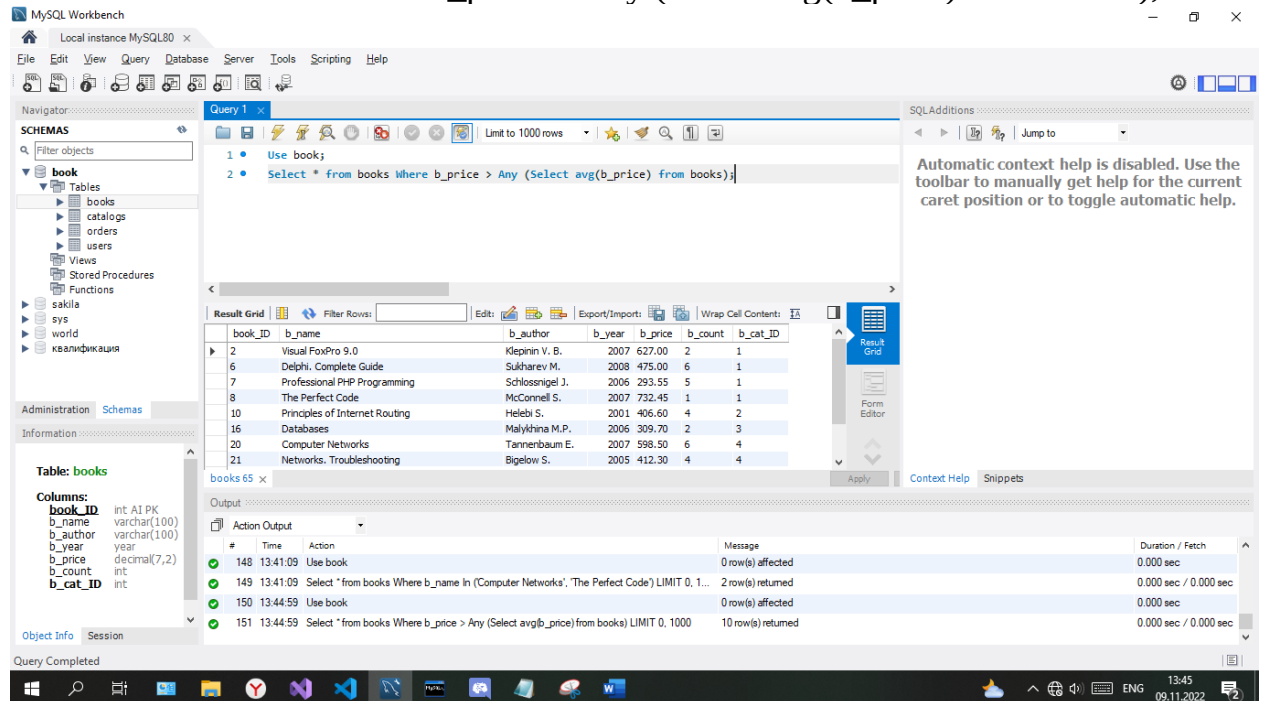
The 'Output' tab is also visible, showing the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
146	13:39:29	Use book	0 row(s) affected	0.000 sec
147	13:39:29	Select cat_name from catalogs Where cat_ID Not In (Select b_cat_ID from books where books.b_cat_ID=catalogs.cat_ID)	1 row(s) returned	0.000 sec / 0.000 sec
148	13:41:09	Use book	0 row(s) affected	0.000 sec
149	13:41:09	Select * from books Where b_name In ('Computer Networks', 'The Perfect Code') LIMIT 0, 1...	2 row(s) returned	0.000 sec / 0.000 sec

2.4 Вывести все книги, цена которых выше средней

Use book;

Select * from books Where b_price > Any (Select avg(b_price) from books);



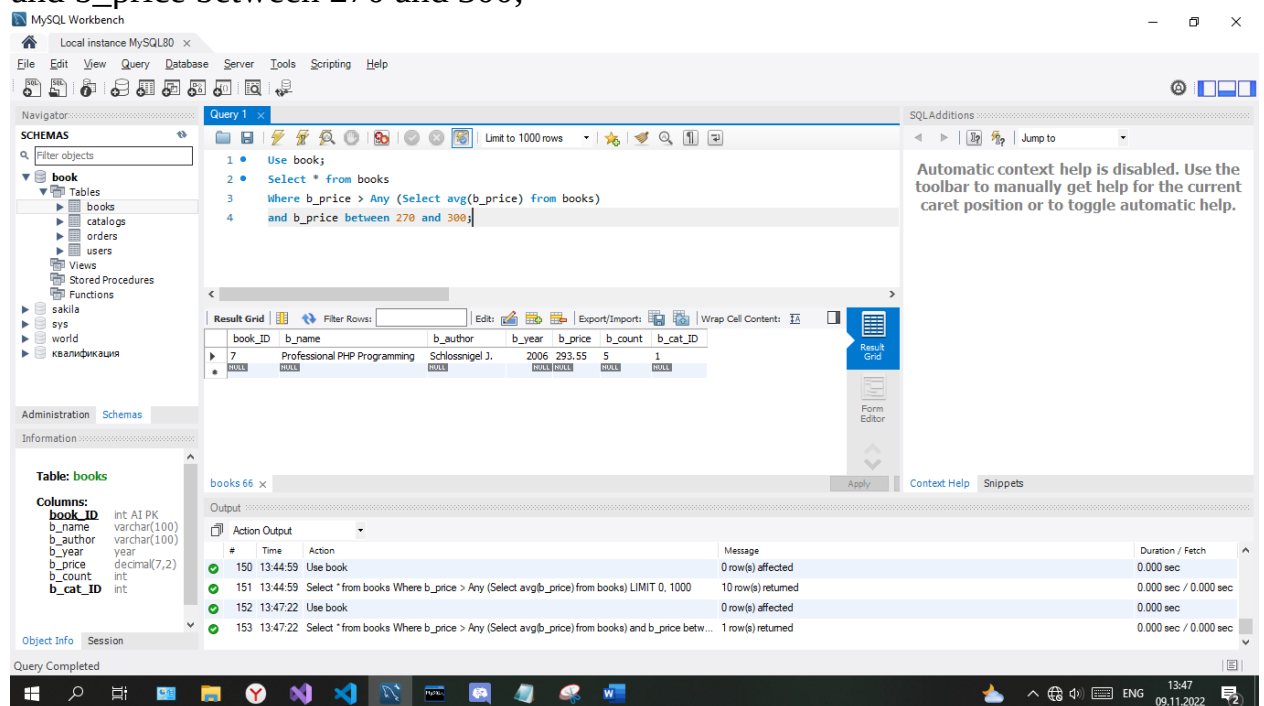
2.5 Вывести все книги, цена которых выше средней и находится в диапазоне между 270р и 300 р (использовать between)

Use book;

Select * from books

Where b_price > Any (Select avg(b_price) from books)

and b_price between 270 and 300;

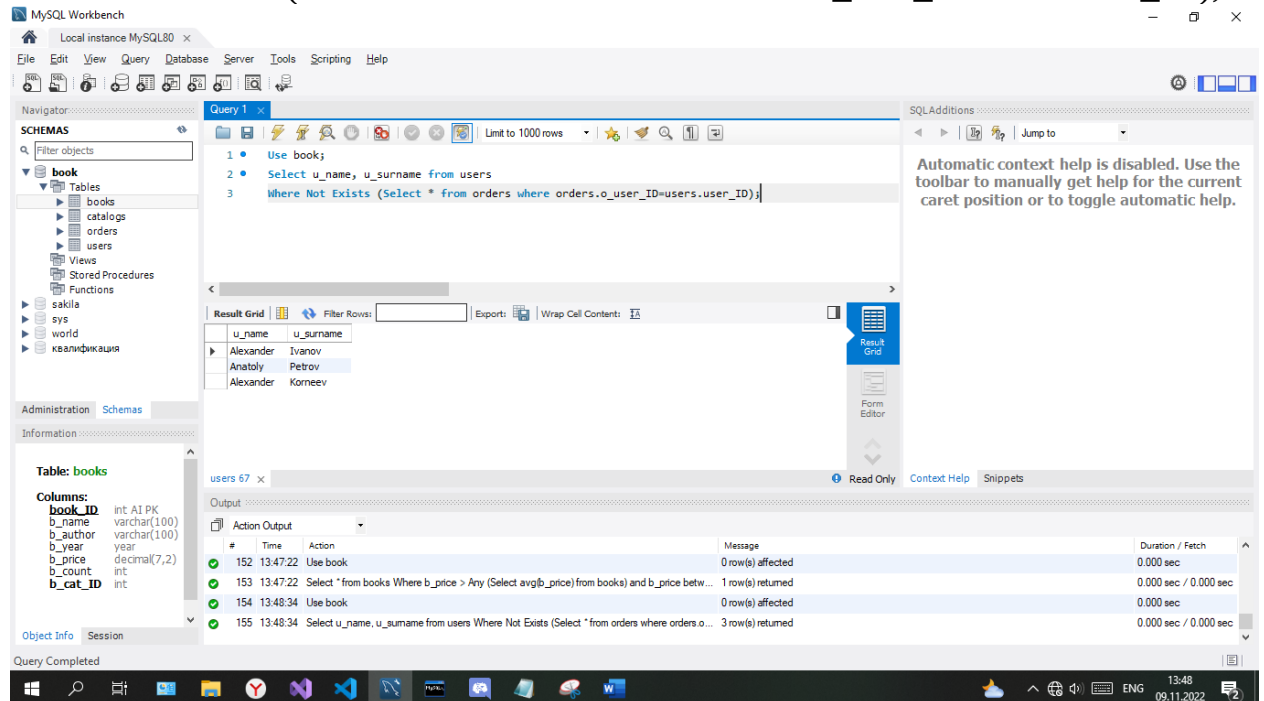


2.6 Показать всех пользователей, никогда не заказывавших книги

Use book;

Select u_name, u_surname from users

Where Not Exists (Select * from orders where orders.o_user_ID=users.user_ID);



Задание 3

Ответьте на контрольные вопросы:

- 1 Чем обусловлено применение LEFT (RIGHT) JOIN в запросах?
(отличие INNER JOIN от LEFT(RIGHT) JOIN)
Inner Join – возвращает данные, присутствующие в обеих таблицах
Left Join – возвращает данные из левой таблицы и совпадающие данные из правой таблицы
- 2 В чем смысл применения вложенных запросов?
Позволяет использовать результат, возвращаемый одним запросом, в другом запросе.
- 3 Какое требование накладывается на применение операторов ANY и ALL?
Использование оператора сравнения
- 4 В каких случаях применяются ключевые слова EXISTS и NOT EXISTS?
Для проверки, содержит ли результирующая таблица хотя бы одну строку